

ArchSync: Evidence-Backed Conformance Checking for TypeScript Service Architectures

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Passing tests do not establish that a service change conforms to its intended architecture. ArchSync links a machine-readable architectural contract to selected HTTP, PostgreSQL, Redis and AMQP observations in TypeScript, retains source evidence, and maps introduced findings to PASS, BLOCK or REVIEW at a Git change. We evaluate this deterministic prototype using co-developed development datasets: 20 isolated architecture patches, 40 annotated detector signals, and an incremental replay of those same patches. A separately reported hardening campaign checks a finite rule domain against a matrix oracle and exercises adversarial source and cache behavior. The controlled observations support contract coherence and regression reliability. They do not establish representative accuracy, comparative superiority or developer productivity; independently labeled repositories and a capability-matched external comparison remain necessary. The contribution is an auditable integration of service-level observations, explicit rules and change-time decisions, together with evidence that exposes its reliability boundaries.

CCS Concepts: • **Software and its engineering** → **Software architecture**; *Software verification*.

Additional Key Words and Phrases: architecture conformance, service-oriented systems, static analysis, architecture drift, reproducibility

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1 INTRODUCTION

Service-oriented systems can preserve functional behavior while introducing a forbidden dependency, losing a required interaction, or adding infrastructure that needs a design decision. These changes concern different architectural judgments. A useful check must connect an observed interaction to the declared contract, identify supporting source evidence, and distinguish a violation from an evolution. ArchSync addresses this information-extraction and decision-support problem with deterministic rules; no learned model or AI effectiveness claim is required.

Our contribution is an executable integration: (1) a typed expected/observed service graph with explicit constraints; (2) provenance-aware extraction of selected TypeScript interactions and evidence; and (3) a Git-diff gate that reports introduced violations separately from evolution. Model comparison and dependency constraints are established ideas. Our empirical contribution is a bounded, reproducible verification of this integration, including failure-first hardening rather than a claim of new general-purpose program analysis.

We ask whether the prototype reconstructs declared graph changes and detector signals (RQ1), classifies controlled changes correctly (RQ2), supplies the declared evidence locations (RQ3), and preserves results under replay and incremental execution (RQ4). These are development-contract questions, not estimates of unseen-project accuracy.

2 BACKGROUND AND RELATED WORK

This scoped narrative synthesis establishes context for ArchSync’s design; it is not a systematic literature review. Sources were selected purposively from the retained keyword-search corpus for architecture conformance, drift detection, and static-analysis reliability, emphasizing 2022–2025. Four earlier works supply concept origins. Selection and interpretation may omit relevant approaches; coverage completeness and absence-of-prior-work claims are outside scope.

Erosion and reconstruction. A mapping of architecture erosion describes its manifestations and consequences [13]. Code-review studies identify symptoms in OpenStack and later OpenStack/Qt projects [12, 14]; their shared project contexts are not independent replications. Practitioner evidence identifies documentation, responsibilities, and shared practices as ways to manage drift [3]. Reconstruction taxonomies organize recovery strategies [7, 19], while a microservice mapping compares tool purposes and availability [4]. View-based analysis compares intended and reconstructed specifications in an industrial case [23]. ArchSync restricts its observation boundary to explicit service and infrastructure relationships.

Conformance. Reflexion Models, static conformance comparisons, and DCL provide the origins of model comparison and allowed/forbidden dependency rules [10, 17, 22]; neither concept is new here. CALint applies dependency-rule checks to Python [6]. ArchLintor is a close TypeScript comparison, detecting module-interaction violations and supporting CI/CD [20]. ArchSync’s typed service topology differs from that module boundary; no head-to-head accuracy result is claimed. IaC metrics address deployment coupling [18], and 3Erefactor searches for executable inconsistency-reducing repairs [15]. LLM-generated architectural tests distinguish execution success from correct rule expression [16]. ArchSync executes explicit rules deterministically, but that choice alone implies neither accuracy nor superiority.

Change-time feedback. Commit-level prediction studies architectural change [25]; repository mining tracks microservice evolution [8]; industrial conformance visualization supplies practitioner evidence [24]. A predicted or observed change need not violate intended architecture. ArchSync instead reports findings introduced at a Git head, preserving separate violation and evolution decisions.

Evaluation reliability. Replication support remains an evaluation concern [11]. Decomposition surveys cover an adjacent problem [1], whereas a microservice-recovery comparison executes nine tools and reports individual and combined recovery performance [21]. That study supplies an external-comparison precedent, not scores directly comparable with ArchSync. Static visual models and multiple dependency models motivate future multi-source work [2, 5]. Interrogation testing exposes both precision and soundness failures [9], motivating our explicit hard negatives. These sources establish comparison dimensions rather than exhaustive novelty or representative accuracy.

3 EXECUTABLE CONTRACT AND CHANGE-TIME ANALYSIS

Let $M = (V, E, R)$ be the expected architecture, with components V , directed typed relationships E , and rules R . The observed graph O is reconstructed from supported source constructs. Component identifiers and metadata from M condition reconstruction; this is not architecture recovery without prior structure. Edges collapse repeated observations with the same source, target and kind, while retaining their file/line evidence.

A deny rule rejects matching edges. An allow rule restricts destinations of the selected outgoing relations. A require rule demands a matching direct edge for each selected source; require-path demands a nonempty directed path,

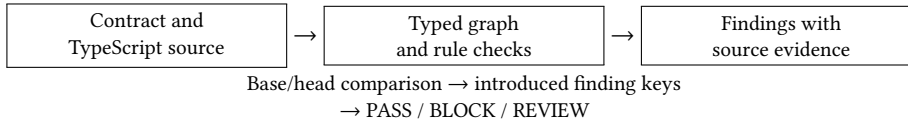


Fig. 1. Source observations and explicit contracts feed a change-time decision. The model remains under separate human control.

optionally restricted to one edge type. An observed rule violation takes classification precedence; otherwise a graph difference is evolution, and an unchanged graph is no-impact. The analyzer never rewrites the expected model.

The TypeScript adapter recognizes selected global fetch calls and imported PostgreSQL, Redis and AMQP client operations. Import provenance distinguishes package clients from method-name lookalikes. HTTP targets use host identity rather than port/path identity; AMQP broker reachability does not demonstrate routing compatibility. Unsupported wrappers, unresolved dynamic endpoints and inter-file flows can remain invisible. Evidence scores are heuristics, not calibrated probabilities.

For a base/head pair, the gate reconstructs the base, analyzes affected source components and merges retained observations with new ones. Full-head scanning supplies a consistency oracle. New violation keys produce BLOCK, new evolution keys produce REVIEW, and no introduced finding produces PASS. Existing findings remain visible. Keys operate at relationship level, so an additional offending call on an existing edge need not introduce a new key. PASS therefore means no newly recognized gated finding, not complete architectural safety. REVIEW requires a separate architecture decision; unchanged inputs remain REVIEW after a human rejection. Repository governance is not implemented by the classification alone.

Call-site evidence points to observed code. A missing required relation has no absent call to localize, so it receives a deterministic component anchor. These evidence types are scored separately. Cache identity binds the baseline and configuration; later hardening also separates repository scope and analyzer version and checks payload integrity. A checksum protects against accidental corruption, not a writer able to replace both payload and checksum.

Worked controlled example. In D1 case 06, rule ARCH-001 forbids a direct relationship from frontend to payment-service. The patch adds `payDirectly` and a fetch call to the payment endpoint. Reconstruction adds one HTTP edge between those components; the deny rule produces an error finding with evidence at `frontend/src/app.ts`, line 14 of the patched file. Because the finding is introduced relative to the base, the Git gate returns BLOCK. This is the frozen case’s declared behavior, not a new production observation. A new relation without a violated rule instead needs REVIEW; the tool does not infer that the architectural change is approved.

4 HISTORICAL CONTROLLED EVALUATION

Objects and independence. D1 is a synthetic Order Platform baseline with five components and five relationships. Its 20 isolated patches comprise nine no-impact changes, seven violations and four evolutions. Topology labels preceded integration, but executable patches and precise evidence locations co-evolved with the analyzer. D2 has four positive and four hard-negative signals for each of five detector groups: fetch, PostgreSQL, Redis, AMQP publish and AMQP consume. It was constructed from v0.1 weaknesses and informed v0.2. Neither dataset is a holdout. P3 replays D1 through Git; it is not a third independent dataset.

Procedure and units. D1 validates hashes, patch applicability and expected changes, then runs the baseline and each isolated patch twice (42 executions). D2 runs both pinned versions twice (four executions). P3 creates fresh baseline repositories, applies one patch, performs cold and warm checks, and separately scans the full head. Results are

Table 1. Historical development-contract results; repeated baseline structure is not independent evidence.

Measure	Agreement/count
D1 changed nodes / edges	5/5; 12/12
D1 classifications / violation rule sets	20/20; 7/7
Call-site cases / missing-relation anchors	9/9; 2/2
P3 decisions / exact deltas	20/20; 20/20
P3 incremental/full scan / cold-warm replay	20/20; 20/20
P3 parsed / head file instances	57/189

compared after excluding elapsed time and cache-hit state. The 20 cold/warm pairs plus full scans require 80 analyzer calls. D1 items are graph nodes/edges; D2 items are annotated detector–edge–file–line signals, which can share graph edges. Evidence matches are case-level, not recall over all call sites.

Results. Table 1 reports the meaningful changed-item and case denominators. Full-graph totals also matched 105 nodes and 108 edges, but repeatedly include baseline structure. On D2, v0.1 yielded TP=18, FP=4, FN=2 and TN=16; v0.2 yielded TP=20, FP=0, FN=0 and TN=20. The four false positives were local PostgreSQL/AMQP-shaped receivers; the two misses were Redis operations. These targeted repairs are within-tool regression evidence, not external comparative improvement. Duplicate normalized output matched, and P3 cache miss/hit results agreed in all 20 cases. Incremental/full agreement can preserve the same semantic omission in both modes.

The original Windows recording used Node.js 22.16.0 and an Intel Core i7-13650HX. Cold/warm nearest-rank p50 values were 518.51/242.62 ms and p95 values 531.05/249.30 ms. They combine parsing, Git, cache I/O and reporting; RAM, power/load and filesystem-cache state were unrecorded. These are descriptive observations, not causal speedup or scalability evidence. Historical functional replays passed Linux and Windows; current local hardening below does not constitute a new cross-platform study.

5 SEPARATE TECHNICAL RELIABILITY EXTENSION

The subsequent 2026-09-26 campaign targets local development defects and preserves original failures. It does not revise the preceding observations or their pinned analyzer packages.

Method provenance. OpenAI Codex with delegated coding agents assisted in constructing the finite oracle, adversarial inputs and mutation harness, modifying the implementation, executing local checks and inspecting outputs. Reported counts come from retained executable test reports, not generated estimates. Test refinement after inspecting failures is disclosed below; the checks are development evidence. Cross-agent inspection is automated code review, not independent human annotation or adjudication. No human validation signoff is asserted for this extension.

Finite rule checking. A separately implemented adjacency-matrix and transitive-closure oracle checks all 4,096 directed graphs on three fixed nodes, two edge types and 12 non-self edge slots. Each graph is evaluated against 192 combinations of deny/allow/require/require-path, exact/wildcard selectors and type constraints: 786,432 rule–graph evaluations. These are finite combinations, not independent samples. Additional checks apply ordering and consistent renaming on 241 graph states and inverse addition/removal deltas on all 4,096. Self-loop inputs, larger graphs, arbitrary wildcard fragments, extraction and component-metadata changes are outside this domain.

Test sensitivity. Eight declared implementation mutations target rule enforcement, path traversal and classification. The initial suite killed seven; violation demotion survived because classification was not asserted. Adding the missing

assertion produced eight kills, no survivors and no execution errors in the final audit. Both rounds are retained. This is test-development evidence against a small fault model, not a whole-program mutation score or independent validation.

Extraction and incremental faults. Adversarial source tests exposed type-only imports treated as runtime clients, protocol mismatches, shadowed bindings and mistaken URL expressions. Binding symbols per file separates local names from package/global names. Later checks exposed wrong-host string concatenation and stale provenance after direct reassignment. Static string composition now resolves the complete URL; direct assignment conservatively invalidates the binding throughout its file, including earlier legitimate uses. Unresolved composition emits no relationship. The repair is not flow-sensitive soundness.

Incremental checks exposed missed unusual/ignored filenames, nested-component evidence loss, cross-project cache reuse and corrupt-cache behavior. Exact source ownership, Git path handling, repository/version-bound cache identity and payload checks address the reproduced cases. The extension records failing inputs, repaired outputs and remaining limitations separately. Its full local suites passed 126 Core and 301 Guardian tests on Node.js 22.16.0/macOS arm64; those totals describe regression execution only. Two inert compiler-host callbacks are excluded from Guardian’s configured coverage. The current analyzer is unreleased semantics version 0.3; its passing reuse of all 20 D1 cases is still development evidence.

6 VALIDITY AND IMPLICATIONS

Construct and internal validity. A typed structural view does not measure responsibilities, runtime quality or usefulness. Co-developed labels and implementation share assumptions. A separately coded oracle may still share a mistaken specification. D2 specificity covers explicit hard negatives, not all possible nonedges. File/line agreement measures one declared case location, not evidence completeness or ranking. Perfect replay can reproduce an incorrect inference.

External and comparative validity. One synthetic system and targeted fixtures cannot establish representative TypeScript accuracy. Dynamic templates, alternative fallbacks, property and destructuring writes, wrappers and inter-file flows remain limitations. No accepted external-tool comparison or independently labeled real-repository holdout exists. Module-import dependencies and service calls require a nonempty common abstraction before comparison; unsupported relations must not become comparator false negatives. Narrative source selection can omit relevant work and does not establish exhaustive novelty. Missing full texts remain identified in the review register; no broader review is implied.

Conclusion validity. The small changed-item counts and finite graph enumeration do not justify population confidence intervals or significance tests. Coverage and mutant kills measure tested implementation behavior, not production accuracy. The next study must freeze repository selection and tools before predictions, label the entire declared source scope (including files without predictions), retain annotation disagreements, and report per-repository errors, supported-vocabulary coverage and failures. A paired, controlled workload study is needed for performance; practitioner observations are needed for false-block burden and review effort.

7 CONCLUSION

ArchSync combines service-level extraction, explicit conformance rules and evidence-backed Git decisions. Controlled datasets establish agreement with development contracts; the separate hardening campaign finds and repairs concrete reliability faults while exposing residual limitations. The strongest supported result is auditable engineering coherence. Independent repositories, compatible baselines and practitioner evaluation remain necessary before broader accuracy or usefulness claims.

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A REPRODUCTION AND INTERPRETATION DETAILS

This appendix supplies operational detail for the single-column review profile. It is supplied as a separate supplement with the compact profile; acceptance of supplementary uploads is not established by the public instructions. Its purpose is to make the development evidence inspectable, not to add independent observations.

A.1 Keep historical and current evidence separate

Historical D1/D2/P3 observations are reproduced from their pinned benchmark inputs and packaged analyzers. The reporting derivation uses benchmark commit 24d63ebf2fc3075a1d64f1eaff38cdc0b7f586fb. Its ground truth, Phase 2 and Phase 3 records have individual hashes. Rerunning a current analyzer on that same corpus answers a different question: whether the updated implementation preserves known development behavior. The latter cannot silently replace the former in a before/after table.

The artifact therefore distinguishes original packages, current source snapshots, current distribution builds, first failing runs, repaired runs and unchanged historical result inputs. Current source and package hashes identify the evaluated candidate; an unreleased version string alone does not identify its bytes. The accompanying manifest maps each new manuscript count to its raw receipt. A successful package installation is an operational compatibility check, not an independent assessment of architectural correctness.

The historical evidence-location decomposition is explicitly post-hoc. Nine finding-bearing cases concern call sites; cases 08 and 17 concern deterministic anchors for missing edge/path requirements. Case 15 contains two violated rules but counts once. A match in these finite cases cannot establish that all useful evidence was found, correctly ranked or understood by a developer.

A.2 Finite oracle construction

For three nodes, six ordered non-self node pairs and two edge types give 12 binary edge slots, hence $2^{12} = 4096$ graphs. For each graph, an adjacency matrix is built independently for all edges and for each individual edge type. Direct requirements query that matrix; path requirements query its transitive closure. The diagonal is not initially true: a required path must contain an edge. A cycle may nevertheless establish a nonempty path back to its starting node.

For each of four rule kinds, the source and target independently range over three exact identifiers and a full wildcard, and the edge-type restriction ranges over unrestricted and the two types. Thus $4 \times 4 \times 4 \times 3 = 192$ rule configurations are checked for every graph. The oracle compares per-rule violation counts, aggregate violations and classification. It imports no production graph-building, selector-matching or traversal helper. This structural separation reduces direct implementation reuse but does not establish independent authorship or correctness of the shared rule interpretation.

Ordering and consistent-renaming checks use masks 0 through 4080 in steps of 17, giving 241 graph states. Inverse-delta checks compare every graph with the empty-edge graph. They verify that added edges in one direction are removed edges in the other, and that edge changes without rules classify as evolution. These checks do not enumerate arbitrary pairs of graphs, metadata changes, arbitrary wildcard fragments or source-language programs. The chosen domain is exhaustive only within its declared bounds.

A.3 Mutation procedure and retained failures

The mutation harness copies the relevant source and test into a temporary isolated workspace, changes one declared expression at a time, and invokes the same test suite with a fixed timeout. Production source remains unchanged by this procedure. Every run retains the replacement text, original and mutated source hashes, test hash, command output, exit state and native report. A passing unmodified baseline is required before mutant outcomes are interpreted.

Mutant states distinguish test passes, assertion failures and execution errors. The eight faults disable deny enforcement, invert allow logic, invert direct-require satisfaction, ignore path edge types, prevent traversal beyond one edge, miss path target detection, demote violation, or demote evolution. The scope is intentionally narrow. A compile error or timeout is not silently counted as an assertion kill, and no claim about all equivalent mutants is made.

The original surviving classification mutation matters: it demonstrated that correct finding counts alone did not guarantee the test checked the returned decision. After the assertion was added, the same fault was detected. Reporting only the final perfect score would hide that development intervention. Both versions of the test evidence remain available, so reviewers can distinguish test strengthening from confirmation on an untouched sample.

A.4 Detector repairs and residual uncertainty

The first adversarial set contains 18 cases: 15 failed before intervention and all 18 passed after it. Positive controls check aliases, captured resources and an unshadowed global fetch alongside local bindings. Type-only import tests deliberately include source that a TypeScript compilation gate would reject. They test the analyzer’s robustness when parsing source without that gate, not the prevalence of such constructs in real projects.

A later intervention added eight cases for concatenation and reassignment: seven failed in the cumulative 26-case pre-repair suite. Three further boundary controls were added after repair, yielding 29 passing adversarial cases. These before/after suites therefore have different total sizes, and a naive percentage comparison would be misleading. An old positive fixture concatenated a suffix into an invalid URL port; the fixture was corrected to a valid path suffix and the invalid-port input retained as a negative regression.

Direct-assignment invalidation is conservative and file-wide. It can suppress a genuine call that appears before reassignment; it does not implement statement-order data flow. Property mutation, destructuring assignment, imported constructor rebinding, dynamically built templates and alternative fallback destinations remain separate risks. When no edge is emitted, the tool does not prove that no runtime relationship exists. Confidence constants attached to emitted evidence cannot resolve this missing-observation uncertainty.

A.5 Incremental and cache experiment boundary

The incremental suite creates temporary Git repositories and checks additions, modifications, deletions, renames within and between components, and complete-component deletion. Cold and warm results are compared with fresh full scans. The initial nine adversarial cases all failed on the original implementation; a later ignored-source case exposed a further false PASS. Before and after receipts preserve those observations.

Most differential cases compare classification, decisions, finding count and scanned-file count. Cold/warm comparisons additionally inspect architecture deltas and introduced/resolved findings. The nested-component case compares the complete observed graph and evidence. This distinction is important: aggregate count agreement alone does not prove identity of every graph edge. The suite does not constitute a proof over all Git configurations or filesystem layouts.

The baseline cache now separates project-relative roots, architecture identity and analyzer semantics. A payload hash rejects accidentally changed JSON; concurrent writers use distinct temporary publication paths. Neither mechanism authenticates a malicious cache writer. Git inventory and scanning are not an atomic snapshot, so the worktree must remain stable during a check. Ignored source outside excluded build/dependency directories is included because full scanning would read it; the gate must not ignore a relationship merely because Git does.

A.6 Independent follow-up design

The technical extension specifies a future repository frame selected before prediction inspection, with lineage-aware development/pilot/holdout separation. Two independent human reviewers must enumerate supported relations across the complete declared source scope, seal their separate labels, and preserve disagreements before adjudication. Checking emitted predictions alone can estimate precision but cannot establish recall. Supported-vocabulary coverage, execution failures and unresolved labels require separate accounting.

A comparator must share a nonempty, frozen observation unit with ArchSync. An import dependency and a remote service interaction are not interchangeable merely because both can be drawn as graph edges. Performance experiments must pair the same workloads and machines, control the graph-cache condition, separate priming from timed execution and retain failures. Operating-system disk cache is not equivalent to the ArchSync graph cache. These protocols are proposed technical designs, not completed studies or evidence of independent human participation.